ACADEMIC PROJECT DEFENSE • TOTAL QUALITY MANAGEMENT

Total Quality Management (TQM) in Healthcare Systems

Case Study: AegisPharm Enterprise — Fail-Safe Clinical Pharmacy Architecture & Continuous Quality Assurance

PRESENTER INFORMATION:

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ACADEMIC CONTEXT:

🏛️ **Amrapali University**

Course: Total Quality Management (TQM)
Healthcare Software Engineering & Operations

PROBLEM STATEMENT & CLINICAL CONTEXT

The Quality Crisis: Why TQM is Critical in Pharmaceutical Healthcare

🚨 DEFECT #1: MEDICATION ERRORS

• Adverse Drug Events (ADEs): Dispensing medications contraindicated by patient allergy histories (e.g. Penicillin, NSAIDs).

• Harmful Interactions: Overlooking severe drug-drug interactions (e.g., Warfarin + Augmentin).

• Handwriting & Transcription: Manual illegible prescription misinterpretations leading to toxic dosages.

• TQM Remedy: Automated Clinical Decision Support System (CDSS) Poka-Yoke gate.

❄️ DEFECT #2: THERMAL & EXPIRY LOSS

• Cold-Chain Spoilage: Vaccines, insulin, and biologics degrade irreversibly outside the critical 2.0°C–8.0°C range.

• Dead Stock Capital: Dispensing newer batches first causes older stock to expire on shelves (FIFO vs. FEFO failure).

• Financial Leakage: Millions lost annually in expired, unsellable inventory write-offs.

• TQM Remedy: First-Expiry, First-Out (FEFO) batch picker + continuous 24/7 IoT thermal telemetry.

⚖️ DEFECT #3: COMPLIANCE & SHRINK

• Schedule X & NDPS Violations: Inadequate narcotic logs invite severe regulatory penalties and license revocation.

• Till Variance & Shrinkage: Cash drawer shortages, unverified UPI transfers, and manual till closing errors.

• Audit Trail Absence: Lack of immutable proof when regulatory drug inspectors audit stock.

• TQM Remedy: Dual-PIN Schedule X register + Daily Z-Report financial reconciliation.

THEORETICAL FOUNDATIONS

TQM Conceptual Framework & Translation into Software

1. Customer Focus (Patient Safety)
In healthcare TQM, the customer is the patient. Zero-defect dispensing, allergy prevention, transparent GST billing, and chronic adherence are the supreme quality metrics.

2. Total Employee Involvement
Quality cannot be policed by one manager. AegisPharm empowers 5 distinct roles (Admin, Pharmacist, Cashier, Inventory, Delivery) with specialized workstations and quality responsibility.

3. Process-Centric Approach
Work is governed by standardized clinical pathways. A prescription cannot move to POS billing until it passes the Clinical Verification gate and CDSS allergy check.

4. Continuous Improvement (Kaizen)
Incremental feedback loops: real-time cold-chain alarms, customer refill WhatsApp alerts, and daily register closing ensure continuous operational refinement.

5. Fact-Based Decision Making
Decisions are driven by empirical data: real-time IoT temperature logs, FEFO expiry analytics, and Daily Z-Report cash drawer reconciliation.

TOTAL EMPLOYEE INVOLVEMENT & PROCESS QUALITY

System Architecture: 5 Quality Inspection Gates

Gate 1: Store Admin HQ

Role: Store Administrator

- Role-Based Access Control (RBAC)
- Session presence heartbeat monitor
- Immutable audit trail logging
- Automated SQLite database backups

Gate 2: Clinical Station

Role: Registered Pharmacist

- CDSS allergy shield cross-check
- Drug-drug interaction detector
- Schedule X narcotic PIN sign-off
- Chronic refill adherence tracking

Gate 3: POS Express Billing

Role: Cashier Station

- 1-Click approved Rx import
- Auto FEFO batch selection
- Cash / UPI QR / Card multi-tender
- Automated 5% CGST/SGST tax split

Gate 4: Inventory Hub

Role: Inventory Manager

- Multi-batch FEFO stock ledger
- Goods Receipt Notes (GRN)
- 2.0°C–8.0°C IoT thermal telemetry
- Expiry countdown (<30d, <90d)

Gate 5: Delivery Fleet

Role: Logistics Agent

- Route Kanban status dispatch
- Cold-bag packaging verification
- 4-Digit customer OTP handshake
- Digital Proof of Delivery (e-POD)

FAIL-SAFE QUALITY ENGINEERING

Poka-Yoke: Mistake-Proofing Healthcare Operations

1. CDSS Patient Allergy Shield

POKA-YOKE MECHANISM: AUTOMATED BLOCKING GATE

The cashier or pharmacist cannot bypass or dispense drugs containing penicillin, NSAIDs, or sulfas if the patient EHR contains registered hypersensitivity. Prevents anaphylaxis at the software level.

2. Molecule-Molecule Interaction Guard

POKA-YOKE MECHANISM: CROSS-REFERENCE ALERT

When combining multiple drugs in a prescription (e.g. Warfarin + Augmentin, Glycomet + Alcohol), the Clinical Decision Support System instantly triggers a severe warning modal preventing accidental toxicity.

3. Schedule X Narcotics Dual PIN Sign-Off

POKA-YOKE MECHANISM: MANDATORY DUAL AUTHENTICATION

Narcotic sedatives (e.g. Alprax, Diazepam) cannot be added to the POS cart without doctor registration number validation and a 4-digit pharmacist authorization PIN, eliminating unauthorized diversion.

4. 4-Digit OTP Electronic Proof of Delivery

POKA-YOKE MECHANISM: CRYPTOGRAPHIC HANDSHAKE

A delivery agent cannot mark an order as 'Delivered' simply by clicking a button. The customer must provide a 4-digit SMS OTP, guaranteeing medicines are handed to the rightful verified patient.

CONTINUOUS IMPROVEMENT (KAIZEN)

Deming's PDCA Cycle Applied to Pharmacy Operations

PLAN (Setting Quality Standards)

- Define minimum inventory reorder buffers (≤ 20 units).
- Configure 2.0°C–8.0°C cold-chain thermal thresholds.
- Program chronic patient refill schedules (30/60/90 days).
- Set default cash float drawer balance (₹2,000.00).

DO (Standardized Execution)

- Inward validated supplier stock via Goods Receipt Notes (GRN).
- Intake and review digital doctor e-Prescriptions.
- Dispense FEFO-prioritized batches via Express POS.
- Dispatch orders in cold-insulated delivery bags.

CHECK (Verification & Auditing)

- IoT temperature sensor logs cold-chain readings every 30s.
- CDSS scans incoming prescriptions for allergies & interactions.
- Daily Z-Report compares counted cash vs. system transactions.
- FEFO countdown flags batches expiring in <30d or <90d.

ACT (Corrective & Preventive Action)

- Trigger audio/visual alarms on thermal breaches (>8.0°C).
- 1-Click WhatsApp alerts sent to chronic refill patients.
- Reconcile cash overages/shortages during register closing.
- Create emergency POs for depleted essential medicines.

SHELF-LIFE OPTIMIZATION & BIOLOGICAL INTEGRITY

Inventory Quality: FEFO Scheduling & IoT Cold-Chain

📦 FEFO (FIRST-EXPIRY, FIRST-OUT) ENGINE

Eliminating FIFO Flaws: Traditional FIFO (First-In, First-Out) fails if a distributor delivers newer stock with an earlier expiration date.

Dynamic Batch Sequencing: Algorithms sort medicine batches strictly by earliest calendar expiry date (`expiry_date ASC`).

Color-Coded Visual Expiry Badges:

- Expired (Red): Automatically locked from dispensing.
- Critical < 30 Days (Orange): Targeted for immediate priority fulfillment.
- Attention < 90 Days (Yellow): Monitored on the inventory dashboard.
- Optimal > 90 Days (Green): Standard warehouse stock.

Zero Dead-Stock Waste: Virtually eliminates annual pharmaceutical write-off losses.

❄️ 24/7 COLD-CHAIN IOT TELEMETRY

Strict Biological Range (2.0°C – 8.0°C): Vaccines (e.g. Hepatitis B, MMR), Insulin, and monoclonal antibodies denature outside this threshold.

Continuous Telemetry Streaming: Background sensors log digital temperature readings every 30 seconds into SQLite persistence.

Interactive Circular HUD: Visual circular gauge displays live temperature with color-coded safety states.

Automated Thermal Spikes & Alarms:

- Safe Zone (2.0°C – 8.0°C): Optimal storage integrity.
- Thermal Excursion (> 8.0°C): Triggers audio-visual alarm banner and system audit record.

Cold-Bag Logistics Checklist: Riders must verify cold-gel pack integrity before dispatching temperature-sensitive orders.

STATISTICAL CONTROL & FINANCIAL GOVERNANCE

Fact-Based Decision Making: Daily Z-Report Cash Reconciliation

Gross Sales Revenue
₹ 12,081.00
Real-Time Reconciled Total

GST Liability (5%)
₹ 1,449.72
CGST: 2.5% | SGST: 2.5%

Cash in Drawer
₹ 7,427.50
32 Cash Transactions

Digital Payments
₹ 4,653.50
UPI QR & Card Split

💵 THE CASH DRAWER BALANCING FORMULA

1. Opening Float (Configured Shift Reserve): ₹ 2,000.00
2. (+) Total Cash Sales Processed: ₹ 7,427.50
3. (-) Petty Payouts / Mid-Day Safe Drops: ₹ 0.00
4. (=) Expected Physical Cash in Drawer: ₹ 9,427.50
5. Actual Physical Currency Counted by Cashier: [User Input]
6. Statistical Variance Indicator:
   - 🟢 Perfect Reconciliation (₹ 0.00 Variance)
   - 🔴 Cash Shortage Warning (-₹ X.XX Shrinkage Alert)
   - 🔵 Cash Overage (+₹ X.XX Accounting Surplus)

🖨️ IMMUTABLE REGISTER CLOSE (Z-SLIP)

- Cryptographic Report ID: Auto-generates unique `Z-YYYYMMDD-XXXX` ID.
- SQLite Persistence: Stored permanently in dedicated `z_reports` relational database table.
- Dual Sign-Off: Printable thermal 80mm slip containing Cashier Signature and Store Manager/Auditor verification lines.
- Tally/ERP Compatibility: 1-Click CSV export containing complete shift financial ledger, tender breakdown, and tax splits.
- POS Integration: Accessible to both Store Admin and Cashier POS workstations for shift changeovers.

DEFECT ELIMINATION & METRIC CONTROL

Six Sigma (DMAIC) Framework Alignment

DEFINE

- Define Critical-to-Quality (CTQ) parameters:
  1. Zero medication dispensing errors.
  2. Zero cold-chain thermal excursions.
  3. Zero cash drawer discrepancies.
  4. 100% regulatory narcotics compliance.

MEASURE

- Real-time system telemetry KPIs:
  - Prescription verification turnaround time.
  - Continuous 30s cold-chain temperature logs.
  - End-of-day register cash variance.
  - Inventory velocity & days-to-expiry counters.

ANALYZE

- Root-Cause Analysis via Audit Trail:
  - Identify bottlenecks in prescription verification.
  - Track temperature trends before thermal spikes occur.
  - Audit log search to trace staff accountability.
  - Identify stagnant or dead-stock capital risks.

IMPROVE

- Process Optimization:
  - 1-Click approved Rx import eliminates re-typing.
  - Dynamic UPI QR modal speeds cashier checkout.
  - Automated CGST/SGST tax split calculation.
  - Automated WhatsApp reminder generation.

CONTROL

- Process Standardization & Locks:
  - Role-Based Access Control (RBAC) prevents unauthorized role switching.
  - Tamper-evident SQLite audit logs.
  - Automated database snapshots and backups.

UPSTREAM SUPPLY QUALITY & ISO 9001 STANDARDS

Supplier Quality Management & End-to-End Traceability

🏢 1. Approved Vendor & Supplier Directory

ISO 9001 Clause 8.4 requires strict evaluation of external providers. AegisPharm maintains an authorized suppliers registry with GSTIN numbers, state drug license numbers, designated contact persons, and pre-negotiated credit payment terms (e.g. 30 Days Net). Unregistered vendors cannot supply medicines.

📑 2. Inward Goods Receipt Notes (GRN) Inspection Gate

Incoming stock cannot simply be entered into shelves. The Inventory Manager must record a Goods Receipt Note (GRN) matching the shipment against an authorized Purchase Order (PO). The batch number, manufacturer, received quantity, and expiration date are validated at the loading dock.

🔍 3. Complete Reverse Batch Traceability

If a manufacturer issues a national product recall (e.g., contaminated batch of cough syrup), AegisPharm enables instantaneous reverse lookup. Within 5 seconds, the system identifies which patients bought that exact batch, which doctor prescribed it, and the remaining units in stock.

CUSTOMER EXPERIENCE & DELIVERY FIDELITY

Patient-Centric Quality: Chronic Adherence & e-POD Logistics

🩺 CHRONIC PATIENT ADHERENCE ENGINE

• The Quality Challenge: Over 50% of hypertensive and diabetic patients fail to refill maintenance prescriptions on time, leading to preventable complications.

• Refill Adherence Registry: Tracks dispensation date, pill count, and calculated depletion date.

• Automated Status Classification: Flags patients as 'On Track', 'Upcoming Due', or 'Overdue'.

• 1-Click Care Reminders: Integrated simulation for SMS and WhatsApp refill notifications with personalized medication dosage information.

• Loyalty Point Incentives: Customers accumulate reward points on timely chronic refill pickups, reinforcing positive health behaviors.

🛵 LAST-MILE LOGISTICS & E-POD QUALITY

• Quality in Last-Mile: In medicine delivery, leaving a package unattended or at the wrong doorstep is a serious safety violation.

• Dispatch Kanban Board: Live visual columns for Packaging, Out for Delivery, and Delivered.

• Cold-Bag Packing Checklist: Logistics agents must verify insulated cold-bags and gel ice packs before vehicle dispatch.

• 4-Digit Customer OTP e-POD: Delivery riders cannot mark an order complete without the customer's secret 4-digit code.

• Immutable Audit Timestamp: Records GPS arrival, rider identity, and time of successful handover in SQLite.

DRUGS & COSMETICS ACT (SCHEDULE H1 / X) STANDARDS

Regulatory Compliance as Quality Assurance

Schedule X & H1 Dispensing Register

Under Indian CDSCO and FDA regulations, psychoactive and habit-forming drugs require rigorous tracking. AegisPharm logs Doctor DMC Registration numbers, patient contact details, dispensing pharmacist registration numbers, batch numbers, and exact quantities. Eliminates manual red-book ledger inaccuracies.

Immutable System Security Audit Trail

Every single system operation—user sign-in, prescription rejection, price override, stock adjustment, and patient update—is logged with microsecond timestamps, user IDs, IP/terminal presence, and severity levels (INFO, WARN, CRITICAL). No record can be deleted or backdated.

1-Click Regulatory CSV Exporter

When government Drug Inspectors visit the premises for surprise inspections, Store Admins can generate instant, formatted CSV compliance exports for narcotic registers, batch ledgers, and sales logs in one click, achieving 100% audit readiness.

Automated Database Snapshots & Backups

Ensures business continuity and disaster recovery. Scheduled SQLite database snapshots (`.db` files) safeguard against data loss, power failures, or hardware corruption without requiring costly server infrastructure.

EMPIRICAL QUALITY VALIDATION & TEST RESULTS

Measurable Quality Results & Performance Metrics

0%

Medication Dispensing Errors

Achieved via Poka-Yoke CDSS allergy & interaction barriers.

100%

Cold-Chain Integrity (2–8°C)

Guaranteed through 24/7 continuous telemetry and audio-visual breach alerts.

₹ 0.00

Cash Drawer Variance

Achieved on balanced shifts via Daily Z-Report mathematical reconciliation.

24 / 24

Automated Tests Passing

100% pass rate across end-to-end unit, integration, and security test suites.

📊 TRADITIONAL PHARMACY VS. AEGISPHARM TQM BENCHMARK

Feature / Workflow	Traditional Pharmacy (Manual)	AegisPharm Enterprise (TQM Driven)
Prescription Safety	Manual inspection; prone to doctor handwriting errors	Automated CDSS cross-checks with patient allergy EHR
Inventory Picking	Random or FIFO; older batches expire on shelves	Algorithmic FEFO (First-Expiry, First-Out) guarantee
Cold-Chain Monitoring	Infrequent manual thermometer checks; silent spoilage	Continuous 30s IoT telemetry with automatic alarm alerts
Shift Cash Balancing	Paper ledger estimations; undetected shrinkage	Daily Z-Report mathematical reconciliation & signed Z-slip

PROJECT SUMMARY & STRATEGIC CONCLUSION

TQM is Not an Afterthought — It is the Software Architecture

1. Fail-Safe Patient Safety: Poka-Yoke CDSS barriers guarantee zero preventable clinical drug dispensing errors.
2. Operational & Thermal Excellence: FEFO batch sequencing and IoT telemetry prevent inventory wastage and maintain biological potency.
3. Financial & Regulatory Rigor: Daily Z-Report reconciliation and immutable SQLite audit streams achieve total transparency.
4. Academic & Industry Readiness: Full alignment with Deming's PDCA, Six Sigma DMAIC, and ISO 9001 quality principles.

🙏 THANK YOU FOR YOUR TIME & ATTENTION

Special gratitude to my respected faculty members and project mentors at Amrapali University for their continuous academic guidance and encouragement.

💬 **The floor is now open for questions, feedback, and academic discussion. (Piyush Bhatt — AegisPharm Enterprise)**